

# Comprehensive Mathematical Guide & Formula Compendium

## Chapter 3: Trigonometric Functions

Formulas, Application Methods, Operational Algorithms, and Critical Traps

### 1 Angle Measurement, Conversions, and Arc Relations

#### 1.1 Core Formulas

- **Angle Sign Convention:** Counter-clockwise rotation yields a *positive* angle; clockwise rotation yields a *negative* angle.
- **Sexagesimal Unit Subdivisions:**  $1^\circ = 60'$  (minutes),  $1' = 60''$  (seconds).
- **Radian-Degree Conversion:**

$$\theta_{\text{rad}} = \theta_{\text{deg}} \times \frac{\pi}{180^\circ}, \quad \theta_{\text{deg}} = \theta_{\text{rad}} \times \frac{180^\circ}{\pi}$$
$$1 \text{ radian} = \frac{180^\circ}{\pi} \approx 57^\circ 16' 22'', \quad 1^\circ = \frac{\pi}{180} \approx 0.017453 \text{ radians}$$

- **Arc Length & Central Angle Relation:**

$$l = r\theta \quad (\text{where } \theta \text{ MUST be measured strictly in radians})$$

- **Area of Sector:**  $A = \frac{1}{2}r^2\theta = \frac{1}{2}lr$ .

#### 1.2 When and How to Use

- **Clock Hand Problems:** The minute hand moves through  $360^\circ$  ( $2\pi$  rad) in 60 min ( $6^\circ/\text{min}$  or  $\pi/30$  rad/min). The hour hand moves through  $30^\circ$  ( $\pi/6$  rad) in 60 min ( $0.5^\circ/\text{min}$  or  $\pi/360$  rad/min).
- **Rotational Speed:** Convert revolutions per minute (rpm) to angular velocity  $\omega = 2\pi \times (\text{rpm}/60)$  rad/s.

#### Tricks & Traps: Angle Measurement

1. **Formula Failure ( $l = r\theta$ ):** Never plug angles in degrees into  $l = r\theta$ . Convert  $60^\circ \rightarrow \frac{\pi}{3}$  rad before computing arc length.
2. **Fractional Degree Conversion:** When converting fraction degrees to minutes/seconds, multiply the fractional remainder by 60, take the integer part as minutes, and multiply the remaining fraction by 60 for seconds. Do NOT divide by 100.

### 2 Circular Functions, Domains, Ranges, and Quadrants

#### 2.1 Unit Circle Definition

In a unit circle ( $x^2 + y^2 = 1$ ) with angle  $\theta$  in standard position:

$$\cos \theta = x, \quad \sin \theta = y, \quad \tan \theta = \frac{y}{x} \quad (x \neq 0)$$

## 2.2 Quadrant Signs (ASTC Rule)

- **Quadrant I** ( $0 < \theta < \frac{\pi}{2}$ ): All trigonometric functions are **Positive (All)**.
- **Quadrant II** ( $\frac{\pi}{2} < \theta < \pi$ ): **Sine** and cosecant are positive (**Sine**).
- **Quadrant III** ( $\pi < \theta < \frac{3\pi}{2}$ ): **Tangent** and cotangent are positive (**Tan**).
- **Quadrant IV** ( $\frac{3\pi}{2} < \theta < 2\pi$ ): **Cosine** and secant are positive (**Cos**).

## 2.3 Domain, Range, and Periodicity Table

| Function | Domain                                                            | Range                            | Fundamental Period |
|----------|-------------------------------------------------------------------|----------------------------------|--------------------|
| $\sin x$ | $\mathbb{R}$                                                      | $[-1, 1]$                        | $2\pi$             |
| $\cos x$ | $\mathbb{R}$                                                      | $[-1, 1]$                        | $2\pi$             |
| $\tan x$ | $\mathbb{R} \setminus \{(2n+1)\frac{\pi}{2} : n \in \mathbb{Z}\}$ | $\mathbb{R}$                     | $\pi$              |
| $\cot x$ | $\mathbb{R} \setminus \{n\pi : n \in \mathbb{Z}\}$                | $\mathbb{R}$                     | $\pi$              |
| $\sec x$ | $\mathbb{R} \setminus \{(2n+1)\frac{\pi}{2} : n \in \mathbb{Z}\}$ | $(-\infty, -1] \cup [1, \infty)$ | $2\pi$             |
| $\csc x$ | $\mathbb{R} \setminus \{n\pi : n \in \mathbb{Z}\}$                | $(-\infty, -1] \cup [1, \infty)$ | $2\pi$             |

## 3 Pythagorean Identities and Allied Angle Rules

### 3.1 Fundamental Identities

$$\sin^2 \theta + \cos^2 \theta = 1 \quad 1 + \tan^2 \theta = \sec^2 \theta \quad 1 + \cot^2 \theta = \csc^2 \theta$$

### 3.2 Allied Angle Reduction Algorithm: $f(n\frac{\pi}{2} \pm \theta)$

To transform any expression of the form  $f(n\frac{\pi}{2} \pm \theta)$  where  $n \in \mathbb{Z}$ :

#### 1. Check $n$ (Co-function rule):

- If  $n$  is **even** (e.g.,  $\pi \pm \theta, 2\pi \pm \theta$ ), the function remains **unchanged**:  $\sin \rightarrow \sin, \cos \rightarrow \cos$ , etc.
- If  $n$  is **odd** (e.g.,  $\frac{\pi}{2} \pm \theta, \frac{3\pi}{2} \pm \theta$ ), the function changes to its **co-function**:  $\sin \leftrightarrow \cos, \tan \leftrightarrow \cot, \sec \leftrightarrow \csc$ .

#### 2. Determine the Sign: Assume $\theta$ is an acute angle. Determine the quadrant of $(n\frac{\pi}{2} \pm \theta)$ and apply the sign of the **original function** in that quadrant.

### Negative Angle Identities

$$\sin(-\theta) = -\sin \theta, \quad \cos(-\theta) = \cos \theta, \quad \tan(-\theta) = -\tan \theta, \quad \sec(-\theta) = \sec \theta$$

## Tricks & Traps: Allied Angles & Identities

1. **Sign Source Error:** When evaluating  $\sin\left(\frac{\pi}{2} + \theta\right) = +\cos\theta$  versus  $\cos\left(\frac{\pi}{2} + \theta\right) = -\sin\theta$ , always inspect the sign of the **LEFT-HAND** function in Quadrant II, not the resulting function.
2. **Square Root Fallacy:**  $\sqrt{\sin^2 x} = |\sin x|$ , NOT simply  $\sin x$ . When removing a square root (e.g.,  $\sqrt{1 - \cos^2 x}$ ), you must attach  $\pm$  depending on the quadrant of  $x$ .

## 4 Compound Angle Formulas (Sum and Difference)

### 4.1 Core Identities

$$\begin{aligned}\cos(x+y) &= \cos x \cos y - \sin x \sin y & \cos(x-y) &= \cos x \cos y + \sin x \sin y \\ \sin(x+y) &= \sin x \cos y + \cos x \sin y & \sin(x-y) &= \sin x \cos y - \cos x \sin y \\ \tan(x+y) &= \frac{\tan x + \tan y}{1 - \tan x \tan y} & \tan(x-y) &= \frac{\tan x - \tan y}{1 + \tan x \tan y} \\ \cot(x+y) &= \frac{\cot x \cot y - 1}{\cot y + \cot x} & \cot(x-y) &= \frac{\cot x \cot y + 1}{\cot y - \cot x}\end{aligned}$$

### 4.2 Product of Sum and Difference

$$\begin{aligned}\sin(x+y) \sin(x-y) &= \sin^2 x - \sin^2 y = \cos^2 y - \cos^2 x \\ \cos(x+y) \cos(x-y) &= \cos^2 x - \sin^2 y = \cos^2 y - \sin^2 x\end{aligned}$$

### 4.3 Special Compound Angle Values

- $\sin 15^\circ = \cos 75^\circ = \frac{\sqrt{6}-\sqrt{2}}{4}$ ,  $\cos 15^\circ = \sin 75^\circ = \frac{\sqrt{6}+\sqrt{2}}{4}$
- $\tan 15^\circ = \cot 75^\circ = 2 - \sqrt{3}$ ,  $\tan 75^\circ = \cot 15^\circ = 2 + \sqrt{3}$
- $\tan\left(\frac{\pi}{4} + x\right) = \frac{1+\tan x}{1-\tan x} = \frac{\cos x + \sin x}{\cos x - \sin x}$
- $\tan\left(\frac{\pi}{4} - x\right) = \frac{1-\tan x}{1+\tan x} = \frac{\cos x - \sin x}{\cos x + \sin x}$

## 5 Multiple, Half-Angle, and Power-Reduction Formulas

### 5.1 Double-Angle Formulas

$$\begin{aligned}\sin 2x &= 2 \sin x \cos x = \frac{2 \tan x}{1 + \tan^2 x} \\ \cos 2x &= \cos^2 x - \sin^2 x = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x = \frac{1 - \tan^2 x}{1 + \tan^2 x} \\ \tan 2x &= \frac{2 \tan x}{1 - \tan^2 x}\end{aligned}$$

## 5.2 Power Reduction & Half-Angle Substitutions

### Crucial Simplification Identifiers

$$1 + \cos 2x = 2 \cos^2 x \implies \cos^2 x = \frac{1 + \cos 2x}{2}$$

$$1 - \cos 2x = 2 \sin^2 x \implies \sin^2 x = \frac{1 - \cos 2x}{2}$$

$$\tan \frac{x}{2} = \frac{1 - \cos x}{\sin x} = \frac{\sin x}{1 + \cos x} = \pm \sqrt{\frac{1 - \cos x}{1 + \cos x}}$$

## 5.3 Triple-Angle Formulas

$$\sin 3x = 3 \sin x - 4 \sin^3 x \implies \sin^3 x = \frac{3 \sin x - \sin 3x}{4}$$

$$\cos 3x = 4 \cos^3 x - 3 \cos x \implies \cos^3 x = \frac{3 \cos x + \cos 3x}{4}$$

$$\tan 3x = \frac{3 \tan x - \tan^3 x}{1 - 3 \tan^2 x}$$

## 5.4 Special Sub-Multiple Angle Values

- $\sin 18^\circ = \frac{\sqrt{5}-1}{4}$ ,  $\cos 36^\circ = \frac{\sqrt{5}+1}{4}$
- $\cos 18^\circ = \frac{\sqrt{10+2\sqrt{5}}}{4}$ ,  $\sin 36^\circ = \frac{\sqrt{10-2\sqrt{5}}}{4}$

## 6 Transformation Formulas (Product $\leftrightarrow$ Sum/Difference)

### 6.1 Product to Sum/Difference

$$2 \sin x \cos y = \sin(x+y) + \sin(x-y)$$

$$2 \cos x \sin y = \sin(x+y) - \sin(x-y)$$

$$2 \cos x \cos y = \cos(x+y) + \cos(x-y)$$

$$2 \sin x \sin y = \cos(x-y) - \cos(x+y) \quad \text{(Notice reversed subtraction order!)}$$

### 6.2 Sum/Difference to Product

$$\sin C + \sin D = 2 \sin \left( \frac{C+D}{2} \right) \cos \left( \frac{C-D}{2} \right)$$

$$\sin C - \sin D = 2 \cos \left( \frac{C+D}{2} \right) \sin \left( \frac{C-D}{2} \right)$$

$$\cos C + \cos D = 2 \cos \left( \frac{C+D}{2} \right) \cos \left( \frac{C-D}{2} \right)$$

$$\cos C - \cos D = -2 \sin \left( \frac{C+D}{2} \right) \sin \left( \frac{C-D}{2} \right) = 2 \sin \left( \frac{C+D}{2} \right) \sin \left( \frac{D-C}{2} \right)$$

## 6.3 When and How to Use Transformations

- **Sum to Product:** Use when simplifying ratios or proving conditional identities like  $\sin A + \sin B + \sin C = 4 \cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2}$ .
- **Product to Sum:** Use when integrating products of trig functions or evaluating telescoping product series.

### Tricks & Traps: Transformation Anomalies

1. **The  $\cos C - \cos D$  Trap:** The formula for  $\cos C - \cos D$  contains a negative sign. Writing  $2 \sin \left( \frac{C+D}{2} \right) \sin \left( \frac{C-D}{2} \right)$  without a leading minus sign is one of the most common exam errors.
2. **Telescoping Cosine Product Identity:**

$$\cos A \cdot \cos 2A \cdot \cos 4A \cdots \cos(2^{n-1}A) = \frac{\sin(2^n A)}{2^n \sin A}$$

Multiply and divide by  $2 \sin A$  iteratively to condense the product.

## 7 Trigonometric Equations and General Solutions

### 7.1 General Solution Theorems ( $n \in \mathbb{Z}$ )

| Equation              | General Solution      | Principal Value Range                              |
|-----------------------|-----------------------|----------------------------------------------------|
| $\sin x = \sin y$     | $x = n\pi + (-1)^n y$ | $y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ |
| $\cos x = \cos y$     | $x = 2n\pi \pm y$     | $y \in [0, \pi]$                                   |
| $\tan x = \tan y$     | $x = n\pi + y$        | $y \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ |
| $\sin^2 x = \sin^2 y$ | $x = n\pi \pm y$      | Universal squared form                             |
| $\cos^2 x = \cos^2 y$ | $x = n\pi \pm y$      | Universal squared form                             |
| $\tan^2 x = \tan^2 y$ | $x = n\pi \pm y$      | Universal squared form                             |

### 7.2 Linear Combination Algorithm: $a \sin x + b \cos x = c$

To solve  $a \sin x + b \cos x = c$ :

1. **Check Solvability Condition:** The equation has real solutions **if and only if**:

$$|c| \leq \sqrt{a^2 + b^2}$$

2. **Divide by Hypotenuse:** Divide the entire equation by  $R = \sqrt{a^2 + b^2}$ :

$$\frac{a}{\sqrt{a^2 + b^2}} \sin x + \frac{b}{\sqrt{a^2 + b^2}} \cos x = \frac{c}{\sqrt{a^2 + b^2}}$$

3. **Convert to Single Phase Angle:** Let  $\cos \phi = \frac{a}{\sqrt{a^2 + b^2}}$  and  $\sin \phi = \frac{b}{\sqrt{a^2 + b^2}}$ :

$$\sin(x + \phi) = \frac{c}{\sqrt{a^2 + b^2}}$$

4. Solve using standard general solution formula for sine.

### Tricks & Traps: Solving Equations

1. **Loss of Solutions by Division:** Never divide both sides of an equation by a trigonometric term containing the variable (e.g., dividing  $\sin x \cos x = \sin x$  by  $\sin x$ ). Factor it out:  $\sin x(\cos x - 1) = 0$ . Division loses solutions where the factor equals zero!
2. **Extraneous Roots from Squaring:** Squaring both sides of an equation (e.g.,  $\sin x + \cos x = 1$ ) introduces false roots. You MUST substitute all candidate solutions back into the original equation to verify validity.

## 8 Extreme Values and Extreme Bounds

### 8.1 Linear Combination Bounds

For any real  $x$ :

$$-\sqrt{a^2 + b^2} \leq a \sin x + b \cos x \leq \sqrt{a^2 + b^2}$$

Minimum value is  $-\sqrt{a^2 + b^2}$ , and Maximum value is  $+\sqrt{a^2 + b^2}$ .

### 8.2 Quadratic and Product Extremes

- **Symmetric Product:**  $-\frac{1}{2} \leq \sin x \cos x \leq \frac{1}{2} \implies -\frac{1}{2} \leq \frac{1}{2} \sin 2x \leq \frac{1}{2}$ .
- **Sum of Higher Powers:**

$$\frac{1}{2} \leq \sin^4 x + \cos^4 x = 1 - 2 \sin^2 x \cos^2 x = 1 - \frac{1}{2} \sin^2 2x \leq 1$$

$$\frac{1}{4} \leq \sin^6 x + \cos^6 x = 1 - 3 \sin^2 x \cos^2 x = 1 - \frac{3}{4} \sin^2 2x \leq 1$$